

An explicit upper bound for Voiculescu's constants γ_n

A rigorous partial answer to Problem 1 of arXiv:1810.12497

Autonomous solve-first investigation, lane 12

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Abstract

Voiculescu asks for quantitative bounds on the universal constants γ_n in the formula relating the Lorentz $(n, 1)$ modulus of quasicentral approximation to spectral multiplicity. A direct dyadic construction gives the explicit estimate

$$0 < \gamma_n \leq \frac{2n^n}{12^{n/2}} \quad (n \geq 2).$$

The proof computes every singular value of the finite-rank commutators. We also record the corresponding rectangular scaling and show that increasing the local tensor-polynomial degree cannot improve this construction. No claim of sharpness or new quantitative lower bound is made.

The source problem. For a compact operator X , write

$$|X|_n^- = \sum_{r \geq 1} s_r(X) r^{-1+1/n},$$

where $s_1(X) \geq s_2(X) \geq \dots$ are its singular values. The modulus $k_n^-(\tau)$ is the infimum of the limiting values of $\max_j |[A_m, T_j]|_n^-$ over finite-rank positive contractions $A_m \uparrow I$. For a commuting Hermitian n -tuple τ , the known formula is

$$(k_n^-(\tau))^n = \gamma_n \int_{\mathbb{R}^n} m(s) ds, \tag{1}$$

with $0 < \gamma_n < \infty$. Problem 1 asks for upper and lower bounds for γ_n , $n \geq 2$ [1, p. 21].

1 The dyadic projection calculation

Let $\mathcal{H} = L^2([0, 1]^n)$ and let $\tau = (M_{x_1}, \dots, M_{x_n})$. Its absolutely continuous multiplicity is one on the unit cube and zero elsewhere, so (1) gives

$$\gamma_n = (k_n^-(\tau))^n. \tag{2}$$

For $N = 2^m$, partition the cube into N^n congruent cubes of side $h = N^{-1}$, and let P_N be the orthogonal projection onto functions that are constant on each cube. These projections are finite rank, nested, and increase strongly to I .

Lemma 1 (exact cell block). *For every coordinate j , the operator $[P_N, M_{x_j}]$ has exactly $2N^n$ nonzero singular values, all equal to $1/(\sqrt{12}N)$.*

Proof. Fix one cell Q and put

$$e = |Q|^{-1/2} \mathbf{1}_Q, \quad c = |Q|^{-1} \int_Q x_j dx, \quad w = (x_j - c)e.$$

Then $w \perp e$ and $\|w\|_2 = h/\sqrt{12}$. If $u = w/\|w\|_2$ and P_Q is projection onto $\mathcal{C}e$, then, in the ordered basis (e, u) ,

$$[P_Q, M_{x_j}] = \begin{pmatrix} 0 & h/\sqrt{12} \\ -h/\sqrt{12} & 0 \end{pmatrix}. \quad (3)$$

Indeed, $M_{x_j}e = ce + w$ and $\langle M_{x_j}u, e \rangle = \langle u, M_{x_j}e \rangle = h/\sqrt{12}$. If $v \perp e, u$, then both $P_Q M_{x_j}v$ and $M_{x_j}P_Q v$ vanish, so (3) is the whole nonzero block on $L^2(Q)$. The blocks for the N^n disjoint cells form an orthogonal direct sum, proving the claim. $\square$

Consequently, for every j ,

$$|[P_N, M_{x_j}]|_n^- = \frac{1}{\sqrt{12}N} \sum_{r=1}^{2N^n} r^{-1+1/n}. \quad (4)$$

The elementary power-sum asymptotic $\sum_{r=1}^M r^{-1+1/n} \sim nM^{1/n}$ now gives

$$\lim_{N \rightarrow \infty} |[P_N, M_{x_j}]|_n^- = \frac{n2^{1/n}}{\sqrt{12}}. \quad (5)$$

Theorem 2 (explicit universal upper bound). *For every integer $n \geq 2$,*

$$\boxed{\gamma_n \leq \frac{2n^n}{12^{n/2}}}.$$

Proof. The same P_N works simultaneously for all coordinates, and the quantities in (4) are equal. Thus (5) is an admissible sequence in the definition of $k_n^-(\tau)$, whence $k_n^-(\tau) \leq n2^{1/n}/\sqrt{12}$. Raise this inequality to the n th power and use (2). $\square$

For orientation, the bound is $\gamma_2 \leq 2/3$, $\gamma_3 \leq 3\sqrt{3}/4$, and $\gamma_4 \leq 32/9$. The construction also applies when $n = 1$, but its estimate $1/\sqrt{3}$ is weaker than the known identity $\gamma_1 = 1/\pi$.

2 Scaling and a natural optimization test

Rectangles. On $R = \prod_{j=1}^n [0, L_j]$, divide direction j into N_j intervals and put $M = \prod_j N_j$. The same proof shows that the j th commutator has $2M$ nonzero singular values, all equal to $L_j/(\sqrt{12}N_j)$. Choosing N_j/L_j asymptotically independent of j yields

$$\limsup_j \max_j |[P, M_{x_j}]|_n^- \leq \frac{n2^{1/n}}{\sqrt{12}} |R|^{1/n}.$$

Thus the construction has exactly the volume scaling forced by (1), rather than relying on an accidental symmetry of the unit cube.

Higher local degree. One might try to improve the estimate by projecting on each cell onto tensor polynomials of coordinate degree at most r . The orthonormal shifted Legendre recurrence shows that only the degree- r boundary couples out of this subspace. In each coordinate the commutator therefore has $2(r+1)^{n-1}N^n$ equal nonzero singular values

$$\frac{a_r}{N}, \quad a_r = \frac{r+1}{2\sqrt{(2r+1)(2r+3)}}.$$

Its limiting Lorentz norm is

$$C_{n,r} = n [2(r+1)^{n-1}]^{1/n} a_r. \quad (6)$$

Proposition 3. *For $n \geq 2$ and every integer $r \geq 1$, $C_{n,r} > C_{n,0}$. Hence piecewise constants are optimal within this tensor-polynomial family.*

Proof. After squaring the ratio in (6), the desired inequality is

$$\frac{3(r+1)^{4-2/n}}{(2r+1)(2r+3)} > 1.$$

Since $4 - 2/n \geq 3$ and $3(r+1)^3 - (2r+1)(2r+3) = r(3r^2 + 5r + 2) > 0$, the claim follows. $\square$

Scope and remaining obstruction. The theorem supplies a fully explicit upper bound, which is one side of the source problem. The natural dual approach uses truncated vector Riesz potentials so that a sum of coordinate commutators has a positive rank-one part. Turning that idea into a numerical lower bound requires a sharp, explicit weak-Schatten estimate for those truncated kernels; that step was not obtained. Accordingly, the strict positivity in the abstract is the previously known part of (1), and this packet does not claim sharpness, a new quantitative lower bound, or a full solution of the hybrid problem.

References

- [1] D.-V. Voiculescu, *Commutants mod Normed Ideals*, arXiv:1810.12497 (2018; version 2, 2019).